mathongo

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Designed by
Math Maestro

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JEE Mains 2020 Jan Chapter wise Question Bank

Sequences and Series

Q1

Find greatest value of k for which $49^k + 1$ is factor of $1 + 49 + 49^2 \ldots\ldots(49)^{125}$

k का अधिकतम मान होगा जबकि $1 + 49 + 49^2 \ldots\ldots(49)^{125}$ का एक गुणनखण्ड $49^k + 1$ है–

(1) 63 (2) 65 (3) 2 (4) 5

7th Jan Morning

Sol

(1)

$$\frac{(49)^{126} - 1}{48} = \frac{((49)^{63} + 1)(49^{63} - 1)}{48}$$

02

If sum of 5 consecutive terms of 'an A.P is 25 & product of these terms is 2520. If one of the terms is – 1/2 then the value of greatest term is

यदि समान्तर श्रेणी के 5 क्रमागत पदों का योगफल 25 है तथा इन पदों का गुणनफल 2520 है यदि उनमे से एक पद – 1/2 है तब सबसे बडे पद का मान है–

(1) $\frac{21}{2}$ (2) 16 (3) 5 (4) 7

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Sol

Let terms be (माना कि पद) $a-2d, a-d, a, a+d, a+2d$.

sum योगफल= 25 $\Rightarrow$ $5a = 25 \Rightarrow a = 5$

Product गुणनफल= 2520

$(5-2d)(5-d)5(5+d)(5+2d) = 2520$

$\Rightarrow (25-4d^2)(25-d^2) = 504$

$\Rightarrow 625 - 100d^2 - 25d^2 + 4d^4 = 504$

$\Rightarrow 4d^4 - 125d^2 + 625 - 504 = 0$

$\Rightarrow 4d^4 - 125d^2 + 121 = 0$

$\Rightarrow 4d^4 - 121d^2 - 4d^2 + 121 = 0$

$\Rightarrow (d^2-1)(4d^2-121) = 0$

$\Rightarrow d = \pm 1, \quad d = \pm \frac{11}{2}$

$d = \pm 1$, does not give $\frac{-1}{2}$ as a term

$d = \pm 1$, जो $\frac{-1}{2}$ नहीं देता है

$\therefore d = \frac{11}{2}$

$\therefore$ Largest term अधिकतम पद $= 5 + 2d = 5 + 11 = 16$

03

Let $3 + 4 + 8 + 9 + 13 + 14 + 18 + \ldots\ldots\ldots 40$ terms $= S$. If $S = (102)m$ then $m =$

माना $3 + 4 + 8 + 9 + 13 + 14 + 18 + \ldots\ldots\ldots 40$ पद तक $= S$. यदि $S = (102)m$ तब $m =$

(1) 20 (2) 25 (3) 10 (4) 5

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Sol

(1)

$S = \underbrace{3+4}+\underbrace{8+9}+\underbrace{13+14}+\underbrace{18+19}\ldots\ldots\ldots 40$ terms पद तक

$S = 7 + 17 + 27 + 37 + 47 + \ldots\ldots 20$ terms पद तक

$S_{40} = \frac{20}{2}[2\times 7 + (19)10] = 10[14 + 190] = 10[2040] = (102)(20)$

$\Rightarrow m = 20$

04

$a_1, a_2, a_3 \ldots\ldots a_9$ are in GP where $a_1 < 0$,

$a_1 + a_2 = 4$, $a_3 + a_4 = 16$, if $\sum_{i=1}^{9} a_i = 4\lambda$, then λ is equal to

$a_1, a_2, a_3 \ldots\ldots a_9$ गुणोत्तर श्रेणी में है जहां $a_1 < 0$,

$a_1 + a_2 = 4$, $a_3 + a_4 = 16$, if $\sum_{i=1}^{9} a_i = 4\lambda$, तो λ का मान होगा–

(1) -513 (2) $-\frac{511}{3}$ (3) -171 (4) 171

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Sol

(3)

$a_1 + a_2 = 4 \Rightarrow a_1 + a_1r = 4$ ....(i)

$a_3 + a_4 = 16 \Rightarrow a_1r^2 + a_1r^3 = 16$ ....(ii)

$\frac{1}{r^2} + \frac{1}{4} \Rightarrow r^2 = 4$

$r = \pm 2$

$r = 2$, $a_1(1+2) = 4 \Rightarrow a_1 = \frac{4}{3}$

$r = -2$, $a_1(1-2) = 4 \Rightarrow a_1 = -4$

$$\sum_{i=1}^{a} a_i = \frac{a_1(r^q - 1)}{r-1} = \frac{(-4)((-2)^9 - 1)}{-2-1} = \frac{4}{3}(-513) = 4\lambda$$

$\lambda = -171$

05

If $2^{1-x} + 2^{1+x}$, $f(x)$, $3^x + 3^{-x}$ are in A.P. then minimum value of $f(x)$ is

यदि $2^{1-x} + 2^{1+x}$, $f(x)$, $3^x + 3^{-x}$ समान्तर श्रेणी में है तब $f(x)$ का न्यूनतम मान है–

(1) 1 (2) 2 (3) 3 (4) 4

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Sol

(3)

$$f(x) = \left(\frac{2^{1-x} + 2^{1+x} + 3^x + 3^{-x}}{2}\right)$$

Using AM $\geq$ GM

$f(x) \geq 3$

06

Find the sum $\sum_{k=1}^{20}(1+2+3+......+k)$

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Sol

1540

$= \sum_{k=1}^{20}\frac{k(k+1)}{2}$

$= \frac{1}{2}\sum_{k=1}^{20}(k^2+k)$

$= \frac{1}{2}\left[\frac{20(21)(41)}{6}+\frac{20(21)}{2}\right]$

$= \frac{1}{2}\left[\frac{420\times 41}{6}+\frac{20\times 21}{2}\right]$

$= \frac{1}{2}[2870+210]$

$= 1540$

07

For an A.P. $T_{10} = \frac{1}{20}$; $T_{20} = \frac{1}{10}$ Find sum of first 200 term.

एक समान्तर श्रेणी के लिए $T_{10} = \frac{1}{20}$; $T_{20} = \frac{1}{10}$ प्रथम 200 पदों का योग है।

(1) $201\ \frac{1}{2}$ (2) $101\ \frac{1}{2}$ (3) $301\ \frac{1}{2}$ (4) $100\ \frac{1}{2}$

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Sol

(4)

$T_{10} = \frac{1}{20} = a + 9d$ ....(i)

$T_{20} = \frac{1}{10} = a + 19d$ ....(ii)

$\Rightarrow \quad a = \frac{1}{200}, d = \frac{1}{200} \quad \Rightarrow \quad S_{200} = \frac{200}{2}\left[\frac{2}{200} + \frac{199}{200}\right] = \frac{201}{2} = 100\ \frac{1}{2}$

08

$\sum_{n=1}^{7} \frac{n(n+1)(2n+1)}{4}$ is equal to बराबर है–

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Sol

504

$\frac{1}{4}\left[\sum_{n=1}^{7}(2n^3 + 3n^2 + n)\right]$

$\frac{1}{4}\left[2\left(\frac{7.8}{2}\right)^2 + 3\left(\frac{7.8.15}{6}\right) + \frac{7.8}{2}\right]$

$\frac{1}{4}[2 \times 49 \times 16 + 28 \times 15 + 28]$

$\frac{1}{4}[1568 + 420 + 28] = 504$

09

$2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} 8^{\frac{1}{48}} \cdot$ .... $\infty =$

(1) $\sqrt{2}$ (2) 2 (3) $2^{\frac{1}{4}}$ (4) 1

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Sol

(1)

$$2^{\frac{1}{4}+\frac{2}{16}+\frac{3}{48}+\ldots\ldots\ldots\infty}$$

$$= 2^{\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\ldots\ldots\infty} = \sqrt{2}$$

10

Let a_n is a positive term of a GP and $\sum_{n=1}^{100} a_{2n+1} = 200$, $\sum_{n=1}^{100} a_{2n} = 100$ find $\sum_{n=1}^{200} a_n$

माना a_n गुणोत्तर श्रेढ़ी का धनात्मक पद है तथा $\sum_{n=1}^{100} a_{2n+1} = 200$, $\sum_{n=1}^{100} a_{2n} = 100$, $\sum_{n=1}^{200} a_n$ का मान है–

(1) 300 (2) 150 (3) 175 (4) 225

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Sol

(2)

Let GP is a, ar, ar^2 ....

$$\sum_{n=1}^{100} a_{2n+1} = a_3 + a_5 + \ldots\ldots a_{201} = 200 \Rightarrow \frac{ar^2(r^{200}-1)}{r^2-1} = 200 \qquad \ldots.(1)$$

$$\sum_{n=1}^{100} a_{2n} = a_2 + a_4 + \ldots\ldots\ldots a_{200} = 100 = \frac{ar(r^{200}-1)}{r^2-1} = 100 \qquad \ldots.(2)$$

Form (1) and (2) r = 2

add both

$\Rightarrow a_2 + a_3 + \ldots\ldots\ldots\ldots a_{200} + a_{201} = 300 \quad \Rightarrow \quad r(a_1 + \ldots\ldots\ldots\ldots a_{200}) = 300$

$$\sum_{n=1}^{200} a_n = \frac{300}{r} = 150$$

11

Let A = {x : |x| < 2} and B = {x : |x – 2| ≥ 3} then

माना A = {x : |x| < 2} तथा B = {x : |x – 2| ≥ 3} तब

(1) $A \cap B = [-2, -1]$ (2) $A \cup B = R - (2, 5)$

(3) $A - B = [-1, 2)$ (4) $B - A = R - (-2, 5)$

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Sol

(4)

$A = \{ x : x \in (-2, 2) \}$
$B = \{ x : x \in (-\infty, -1] \cup [5, \infty) \}$
$A \cap B = \{ x : x \in (-2, -1] \}$
$A \cup B = \{ x : x \in (-\infty, 2) \cup [5, \infty) \}$
$A - B = \{ x : x \in (-1, 2) \}$
$B - A = \{ x : x \in (-\infty, -2] \cup [5, \infty) \}$

12

Number of common terms in both sequence 3, 7, 11, …………407 and 2, 9, 16, ……..905 is

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Sol

(14)

First common term = 23
common difference = $7 \times 4 = 28$
Last term ≤ 407
$\Rightarrow \quad 23 + (n-1) \times 28 \leq 407$
$\Rightarrow \quad (n-1) \times 28 \leq 384$
$\Rightarrow \quad n \leq 13.71 + 1$
$\quad n \leq 14.71$
So $\quad n = 14$